MIL-F-60589B(MU)
5 October 1970
SUPERSEDING
MIL-F-60589A(MU)
27 March 1968

MILITARY SPECIFICATION

FUZE, MT, XM711 ASSEMBLY

1. SCOPE

1.1 This specification covers Fuze, MT, XM711 Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATION

Military

MIL-P-60412	- Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes: General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Component Environmental and Performance Test for
MIL-STD-1167	- Ammunition Data Card
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective

FSC 1390

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2.1 (Cont'd)

DRAWINGS

Munitions Command

C8886458	- Box, (Wood) Wirebound, Packing, Ammunition for Artillery Type and Rocket Fuzes
C9214705	- Carton, Packing, Ammunition, for Artillery Type and Rocket Fuzes
C9214704	- Box, Wirebound, Packing for Artillery Type or Rocket Fuzes
F8864492	- Box, Ammunition, Metal for Artillery Type and Rocket Fuzes
D10542845	- Fuze, MT, XM711 Assembly
IEL-10542845	- Index of Inspection Equipment List for Fuze, MT, XM711

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards, and specifications, except as provided in 6.1.

3.2 Function.

3.2.1 Static regulation.-The movement assembly shall be regulated for nine (9) seconds in accordance with drawing and engineering notes applicable thereto. The accumulated error shall not exceed plus or minus 0.23 percent.

3.2.2 Operation.-The movement shall run from fully wound to unwound and release the firing pin while being spun at 23,900 R.P.M. $\pm$ 100 R.P.M.

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3.3 Environmental

3.3.1 Jolt.- The fuze shall comply with MIL-STD-331, Test No. 101

3.3.2 Jumble.- The fuze shall comply with MIL-STD-331, Test No. 102

3.3.3 Forty-Foot Drop.- The fuze shall comply with MIL-STD-331, Test No. 103.

3.3.4 Waterproofness.- The fuze shall comply with MIL-STD-331, Test No. 108, with the exception that the fuze is to be tested under one foot of water without additional pressure, for one hour minimum.

3.3.5 Transportation - Vibration.- Fuze shall comply with MIL-STD-331, Test No. 104.

3.4 Ballistic

3.4.1 Muzzle Action.- The fuze shall initiate detonation of the round from zero (0) to fifty (50) feet of the muzzle of the weapon within the temperature range of minus 45 degrees to plus 45 degrees Fahrenheit (F) when fired as follows:

Weapon	Cartridge	Fuze Setting
M48A1 Tank, 90MM T139 Gun M41 Tube, M2 Recoil Mechanism	XM580E1	MA

3.4.2 Distance-To-Burst.- The fuze shall initiate detonation of the round during flight within the distance range related to the time limits specified in Table I when fired as follows:

Weapon	Cartridge	Fuze Setting
M48A1 Tank, 90MM T139 Gun M41 Tube, M8 Recoil Mechanism	XM580E1	400 & 3000 meters

3.4.3 Low Temperature Function.- The fuze temperature conditioned at minus 40 to minus 50 degrees Fahrenheit (F) shall initiate detonation of the round during flight when fired as specified in 3.4.1 or 3.4.2, as applicable.

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3.4.4 High temperature function.-The fuze temperature conditioned at plus 140 to plus 150 degrees Fahrenheit (F) shall initiate detonation of the round during flight when fired as specified in 3.4.1. or 3.4.2 as applicable.

TABLE I

Weapon	Cartridge	Setting	Upper Limit (Sec)	Lower Limit (Sec)
90MM T139 Gun	XM580E1	400 meters	.452	.148
90MM T139 Gun	XM580E1	3000 meters	4.175	3.825

3.4.5 High and low temperature conditioning.-For the high and low temperature firings only the fuze shall be temperature conditioned. The elapsed time from the conditioning chamber to the firing in the weapon shall not exceed 5 minutes. Reconditioning is not permitted.

3.5 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.5.2 Surface finish.-The requirements for surface finish are as detailed on the applicable drawings.

3.5.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing.

3.5.4 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.5 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The methods and agents used in processing and cleaning shall not adversely affect the finish or contaminate the parts.

3.5.6 Protective finish.-Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Lot formation.-Each inspection lot of fuzes shall be comprised of components (except explosive components) each produced by one manufacturer by one process in accordance with the same drawing and revision and specifications and revisions, as applicable, and assembled on one assembly line. Each explosive component within a lot of assembled fuzes shall be limited to one manufacturer and one interfix number.

4.2 First article sample.

4.2.1 Initial production sample.-The first units of production shall be submitted as an initial production sample. The sample shall consist of the following:

360 (Aggregate) component parts as delineated in 4.2.2

450 (Aggregate) subassemblies as delineated in 4.2.3

102 Fuze assemblies as delineated in 4.2.5 and 4.2.6

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection by the contractor and provisional acceptance at source by the Government, 80 fuze assemblies shall be forwarded to a designated proving ground as specified in the contract. Three hundred sixty (360) component parts, 450 subassemblies and 22 fuze assemblies shall be inspected, as provided herein, at a Government laboratory or such other facility as specified in the contract.

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4.2.2 Components.

4.2.2.1 Twenty (20) of each of the (17) components shall be tested as in 4.3.3.2 (Hardness), and 4.3.3.3 (Load).

4.2.2.2 Twenty (20) of each component (1) shall be tested as in 4.3.3.5 (Bend).

4.2.3 Subassemblies.

4.2.3.1 Twenty-five (25) of each of the (10) subassemblies shall be tested as in 4.3.3.7.1 (Static load).

4.2.3.2 Twenty-five (25) of each of the (7) subassemblies shall be tested as in 4.3.3.7.2 (Torque).

4.2.3.3 Twenty-five (25) of each subassembly (1) shall be tested as in 4.3.3.6 (Push or Pull Test).

4.2.4 Acceptance criteria.-Failure of any component or subassembly to comply with the requirements of the applicable drawing shall be cause for rejection of the initial production sample.

4.2.5 Fuze assemblies.-Twenty-two (22) fuze assemblies shall be subjected to the following MIL-STD tests, with the fuzes set on muzzle action "MA".

TEST	QUANTITY	REQUIREMENT	METHOD
Jolt	4	3.3.1	4.4.16.1
Jumble	4	3.3.2	4.4.16.2
Forty Foot Drop	5	3.3.3	4.4.16.3
Waterproofness	4	3.3.4	4.4.16.4
Transportation-Vibration	5	3.3.5	4.4.16.5

4.2.5.1 Acceptance criteria.-The criteria of the applicable MIL-Standard shall apply as determined by disassembly and examination in accordance with the procedures in 4.4.16. Where the accept/reject decision is based on engineering judgement, concurrence by the responsible Technical Agency must be obtained.

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4.2.6 Ballistics.-Eighty (80) of the initial production sample shall be subjected to the following ballistic test phases:

WEAPON	PHASE	TEMP. CONDITION	QUANTITY	REQUIREMENT	METHOD
90MM	MA	Ambient	20	3.4.1	4.4.17.1
90MM	MA	-45°F	5	3.4.1	4.4.17.3
90MM	MA	+145°F	5	3.4.1	4.4.17.3
90MM	400 meters	Ambient	20	3.4.2	4.4.17.2
90MM	3000 meters	Ambient	20	3.4.2	4.4.17.2
90MM	3000 meters	-45°F	5	3.4.3	4.4.17.3
90MM	3000 meters	+145°F	5	3.4.4	4.4.17.3

4.2.6.1 Acceptance criteria.-The occurrence of any of the following shall be cause for rejection of the initial production sample:

a. More than one prior function in the combined distance function phases (see 6.2.1. and 6.2.2).

b. **Failure of more than one round to function within 50 ft. in the Muzzle Action ambient phase.**

c. Failure of more than one round to function when set for distance or to meet distance requirements when set for "Muzzle Action" in the low temperature phase.

d. Failure of more than one round to function when set for distance or to meet distance requirements when set for "Muzzle Action" in the high temperature phase.

e. Any of the following in the ambient distance function phases:

(1) More than two duds or two outlier - offtimes or one dud and one outlier - offtime in the combined phases (see 6.2.2 and 6.2.3).

(2) Failure to meet the time limits in Table I by an estimated lot percent defective greater than the appropriate value given in Table II for each distance function phase (see 4.4.17.2 and 6.2.4).

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be presented in accordance with MIL-STD-105.

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4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defective basis in accordance with MIL-STD-105. For the classification of defects contained herein, the AQL given for each shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.3.2.1 Plate No. 2, Second Lamina, Drawing F9205021 covering a detail of Drawing D10542845.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Diameter of holes D, 1, 20, 21, 32 and 33 incorrect	Gage
102.	Location of holes D, 1, 20, 21, 32 and 33 incorrect	Gage
103.	Flatness incorrect	Gage
104.	Surface finish of holes D, 1, 20, 21, 32 and 33 improper	Visual
105.	Chamfer on holes D, 1, 20, 21, 32 and 33, missing or incorrect	Visual
106.	Metal burred or damaged	Visual
107.	Surface finish on thrust bearing areas of holes D, 1, 20, 21, 32 and 33, improper	Visual

4.3.2.2 Plate No. 4, F9205023 covering a detail of Drawing 10542845.

CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Diameter of holes D, 17, 21, 33, P and N, incorrect	Gage
102.	Location of holes D, 17, 21, 33, P and N, incorrect	Gage
103.	Flatness, incorrect	Gage
104.	Chamfer on holes D, 17, 21, 33, P and N, incorrect	Visual
105.	Surface finish on thrust bearing areas of holes D, 17, 21, 33, improper	Visual
106.	Burr	Visual
107.	Surface finish of holes D, 17, 21, 33, improper	Visual

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4.3.2.3 Plate No. 5, F9205024 covering a detail of Drawing 10542845.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Flatness, incorrect	Gage
102. Diameter of holes 1, P and N, incorrect	Gage
103. Location of holes 1, P and N, incorrect	Gage
104. Surface finish of hole #1, improper	Visual

4.3.2.4 Plate No. 6, F9205025 covering a detail of Drawing 10542845.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Diameter of hole 32, P and N, incorrect	Gage
102. Location of hole 32, P and N, incorrect	Gage
103. Flatness, incorrect	Gage
104. Chamfer of hole 32, P and N, incorrect	Visual
105. Surface finish on thrust bearing areas of hole 32, improper	Visual
106. Burr	Visual
107. Surface finish of hole 32, improper	Visual

4.3.2.5 Housing, F9205032 covering a detail of Drawing D10542845.

CRITICAL	
None defined	
MAJOR AQL = .25%	
101. Minor diameter, incorrect	Gage
102. Major diameter, incorrect	Gage
103. Width of teeth, incorrect	Gage
104. Angle of teeth, incorrect	Gage
105. Concentricity of minor diameter, incorrect	Gage
106. Lubricant or foreign matter on teeth	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 1.0%	
107. I.D. incorrect	Gage
108. O.D. of lip, incorrect	Gage
109. Eccentricity of I.D. with datum, excessive	Gage
110. Location of locating pin hole, incorrect	Gage
111. Location of adjusting stud hole, incorrect	Gage
112. Diameter of adjusting stud counterbore, incorrect	Gage
113. Location of threaded holes, incorrect	Gage
114. Surface finish of I.D., improper	Visual

4.3.2.6 Arbor, Escapement, Drawing C9205054 covering a detail of Drawing D10542845.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Perpendicularity of lever seat with datum "A" incorrect	Gage
102. Diameter of pivots, incorrect	Gage
103. Length between bearing shoulders incorrect	Gage
104. Symmetry of bosses incorrect	Gage
105. Diameter over bosses incorrect	Gage
106. Eccentricity of bosses incorrect	Gage
107. Metal burred or damaged	Visual
108. Surface finishes on bearing surfaces, incorrect	Visual

4.3.2.7 Plate, Firing Pin Safety, Drawing C9205062 covering a detail of Drawing D10542845.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Surface finish improper (pits, dents, die marks, flats, etc. in the working surface contacting firing arm shaft - to be checked after plating.)	Visual

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4.3.2.8 Screw, Bottom Assembly - C10521153 covering a detail of Drawing D10542845.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	100% Inspection
101. Length of screw shank	Gage
102. Chamfer (.015 max.)	Visual

4.3.2.9 Firing Pin, Drawing C10521154 covering a detail of Drawing D10542845.

CRITICAL	
None defined	
MAJOR	AQL = 1.0%
101. Surface finish on shoulder angle, improper	Visual
102. Diameter of striker point below min.	Gage

4.3.2.10 Bushing Guide, C10533948 covering a detail of Drawing D10542845.

CRITICAL	
None defined	
MAJOR	AQL = 1.0%
101. Finish on bearing surfaces, improper	Visual
102. Postiion of locating surfaces	Gage

4.3.2.11 Closing Plug, Drawing D10535653 covering a detail of Drawing D10542845.

CRITICAL	
None defined	
MAJOR	AQL = 1.0%
101. Location of locating pin hole incorrect	Gage

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4.3.2.12 Gear Segment, Setting-drawing 10535748 covering a detail of drawing D10542845

Defect	Method of Inspection
CRITICAL	
None Defined	
MAJOR AQL = .25%	
101. Depth of teeth, incorrect	Gage
102. Major diameter, incorrect	Gage
103. Width of spaces between teeth, incorrect	Gage
104. Angle of teeth, incorrect	Gage
105. Symmetry of teeth	Gage
106. Lubricant on teeth	Visual
MAJOR AQL = 1.0%	
107. Thickness of tooth rack, incorrect	Gage
108. Overall width, incorrect	Gage
109. Burr	Visual
110. Surface finish, improper	Visual

4.3.2.13 "Gear, Escapement, drawing D10542765 covering a detail of drawing D10542845"

Defect	Method of Inspection
CRITICAL	
None Defined	
MAJOR AQL = 1.0 %	
101. Tooth profile, incorrect	Gage
102. Surface finish, improper	Visual

4.3.2.14 Body, drawing F10542769 covering a detail of drawing D10542845

Defect	Method of Inspection
CRITICAL	
None Defined	
MAJOR AQL = 1.0%	
101. Major diameter of external thread, improper	Gage
102. Pitch diameter of external thread, improper	Gage
103. Length to external seating shoulder, incorrect	Gage
104. External diameter incorrect at forward datum	Gage
105. Location of closing plug locating hole incorrect	Gage

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4.3.2.14 (cont'd)

Defect	Method of Inspection
MAJOR	
106. External diameter incorrect at rear datum	Gage
107. Thickness through web of small threaded cavity, incorrect	Gage
108. Diameter of body assembly screw holes, incorrect	Gage
109. Location of body assembly screw holes, incorrect	Gage
110. Diameter below thread undercut exceeds max	Gage
111. Protective finish improper inadequate or defective	Visual
112. Threads damaged or not full	Visual
113. Marking missing, incorrect or illegible	Visual

4.3.2.15 Centrifugal Weights, D10452809 covering a detail of drawing D10542845

CRITICAL

None Defined

MAJOR AQL = 1.0%

101. Overall length, incorrect	Gage
102. Diameter of hole in bent tab, incorrect	Gage
103. Location of hole in bent tab, incorrect	Gage
104. Flatness, incorrect	Gage
105. Parallelism, incorrect	Gage
106. Burr	Visual
107. Surface finish, improper	Visual

4.3.2.16 Plate no. 1 first lamina, drawing F10542821 covering a detail of drawing D10542845

CRITICAL

None Defined

MAJOR AQL = 1.0%

101. Diameter of holes D, A, B, and 20, incorrect	Gage
102. Location of holes D, A, B, and 20, incorrect	Gage
*103. D hole bridge bent beyond max, (to be checked within .040 min dia shown)	Gage
104. Width of setting gear segment slot, incorrect	Gage
105. Surface finish of holes D and 20, improper	Visual
106. Chamfer on holes D and 20, missing or incorrect	Visual
107. Metal burred or damaged	Visual
108. Surface finish of thrust bearing surface of holes D & 20, improper	Visual

* Inspection to be performed on a 100% basis.

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4.3.2.17 Plate no. 2 first lamina-drawing F10542823, covering a detail of drawing D10542845

Defect	Method of Inspection
CRITICAL	
None Defined	
MAJOR AQL = 1.0%	
101. Flatness incorrect	Gage
102. Diameter of hole #17, 65, 66, and 104, incorrect	Gage
103. Location of hole #17, 65, 66, and 104, incorrect	Gage
104. Surface finish on hole #17, incorrect	Visual
105. Chamfer on #17 hole, incorrect	Visual
106. Metal burred or damaged	Visual
107. Surface finish of thrust bearing area of hole #17, improper	Visual

4.3.2.18 Head, drawing C10543846 covering a detail of drawing D10542845

CRITICAL	
None Defined	
MAJOR AQL = 1.0%	
101. Diameter incorrect at basic length	Gage
102. Eccentricity of diameter at basic length with datum A including perpendicularity with datum B	Gage
103. Length to seating shoulder, incorrect	Gage

4.3.2.19 Cap, lower-drawing F10542828 covering a detail of drawing D10542845

CRITICAL	
1. Location of MA set line, incorrect	Gage
MAJOR AQL = 1.0%	
*101. Cumulative error of meter scale exceeds max	Gage
102. Diameter incorrect at forward basic length	Gage
103. Diameter incorrect at rear basic length	Gage
104. Eccentricity of forward diameter with datum A	Gage
105. Eccentricity of rear diameter at basic length datum A	Gage
106. Marking missing, incorrect or illegible	Visual
107. Protective finish improper, inadequate, or defective	Visual

*To be checked from MA line, 200, 400, 2000, 3000, and 4000 meter lines

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4.3.2.20 Plate no. 7-drawing F10551658 covering a detail of drawing D10542845

Defect	Method of Inspection
CRITICAL	
None Defined	
MAJOR AQL = 1.0%	
101. Angular surfaces, incorrect	Gage
102. Location of holes D, R, T, 3, 92, 93, 150, 151, 152, and 153, incorrect	Gage
103. Diameter of holes D, R, T, 3, 92, 93, 150, 151, 152, and 153, incorrect	Gage
104. Surface finish on bearing surface of holes D and 3, improper	Visual
105. Surface finish of holes D, R, T, 3, 150, 151, 152, and 153, improper	Visual
106. Burr	Visual
107. Surface finish on angular surface, improper	Visual

4.3.2.21 Escapement gear and pinion assembly, drawing C9205045 covering a detail of drawing D10542845

CRITICAL	
None Defined	
MAJOR AQL = 1.0%	
101. Eccentricity of gear with datum	Gage
102. Perpendicularity of gear, incorrect	Gage
103. Burr	Visual

4.3.2.22 Escapement lever assembly-drawing D9205056 covering a detail of drawing D10542845

CRITICAL	
None Defined	
MAJOR AQL = 1.0%	
101. Eccentricity of functional radii with datum A, B	Gage
102. Perpendicularity of functional radii with datum C	Gage
103. Metal burred or damaged	Visual
104. Surface finish improper	Visual

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4.3.2.23 Detonator Holder Assembly, drawing C9205110 covering a detail of drawing D10542845

Defect	Method of Inspection
CRITICAL	
1. Detonator extends above flush	Visual
MAJOR AQL = 1.0%	
101. Detonator inserted incorrectly (brown end not facing cushion)	Visual
102. Extension of detent pivot pins exceed max allowable	Gage
103. Crimping or staking inadequate	Visual
104. Crimping or staking extends above flush of plug surface	Visual
105. Metal burred, damaged	Visual

4.3.2.24 Detonator screw assembly-drawing C9205144, covering a detail of drawing D10542845

CRITICAL	
1. Detonator extends above flush of detonator screw	Visual
MAJOR AQL = 1.0%	
101. Overall length exceeds max allowable	Gage
102. Green end of detonator not at bottom (thread end)	Visual
103. Crimping inadequate	Visual
104. Thread damaged	Visual

4.3.2.25 Firing arm assembly-drawing C9212053 covering a detail of drawing D10542845

CRITICAL	
1. Fracture of upright bend of firing arm (prior to application of solid film)	Visual
2. Firing arm not staked to shaft	Visual
MAJOR AQL = 1.0%	
101. Insecure weight	Manual
102. Metal burred or damaged	Visual
103. Lubricant missing	Visual
104. Step or improper blend for safety plate drop off (Prior to application of solid film)	Visual
105. Thrust bearing surface damaged	Visual

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4.3.2.26 Plate no. 7 assembly-drawing D10533947 covering a detail of drawing D10542845

Defect	Method of Inspection
CRITICAL	
None Defined	
MAJOR AQL = 1.0%	
101. Centrifugal weight hitting walls of slot for centrifugal spring	Manual/Visual
102. Heads of screws above flush of surface A	Gage
103. Raised metal on angular surfaces	Visual
104. Insecure Assembly	Manual
105. Part missing	Visual
106. Burr	Visual

4.3.2.27 Laminated plate no. 1 assembly-drawing D10542811 covering a detail of drawing D10542845

CRITICAL	
None Defined	
MAJOR AQL = 1.0%	
101. Escapement spring positioned incorrectly	Gage
102. Metal burred or damaged	Visual
103. Foreign matter	Visual

4.3.2.28 Movement Sub-Assembly-Drawing F10542817 covering a detail of drawing D10542845

CRITICAL	
1. Firing Pin fires on "not fire" position of firing arm (Check prior to assembly of setback weights)	Gage
MAJOR AQL = 0.40%	
101. Firing pin does not fire on "must fire" position of firing arm	Gage
102. Firing pin protrusion less than min. in fired position	Gage
103. Burrs and irregularities which prevent proper seating of the movement in the fuze body	Visual

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4.3.2.28 (cont'd)

Defect	Method of Inspection
104. Firing arm jammed by C-weight or no. 7 plate (No clearance between firing arm weight and #7 plate which prevents backoff of firing arm - to be checked prior to assembly of timing disc support)	Manual/Visual
105. Plates separated	Visual
106. Staking of screws inadequate	Visual
107. Foreign matter (within the mechanism)	Visual
108. Metal burred, damaged	Visual
109. Protrusions above plate no. 7 in path of centrifugal weights, check prior to assembly of timing disc support assembly. (Inspection to be performed on a 100% basis.)	Manual/Visual

4.3.2.29 Movement Assembly, D10542813 covering a detail of D10542845

CRITICAL	
1. Timing disc bushing loose on knurl of the arbor (to be examined prior to assembly of locknut)	Manual/Visual
2. Timing disc retainer insecure, damaged or missing	Manual/Visual
3. Escapement gear not locked sufficiently by lever pallet (to be checked prior to assembly to body)	Visual
4. Timing disc slot touches firing arm upright when rotated	Manual/Visual

MAJOR AQL = 0.40%	
101. Firing pin striker point damaged	Visual
102. Escapement lever not locked on proper side of pallet lock slide	Visual
103. Improper or incomplete assembly	Visual
104. Foreign Matter	Visual
105. Interference of timing disc support or timing disc retainer with firing arm in the cocked position	Manual/Visual
106. Location of timing disc retainer	Gage
107. Total height exceeds maximum	Gage
108. Escapement spring broken, deformed or unseated	Visual
109. Setback weight adjacent to firing arm assemlby jammed.	Manual/Visual
110. Protrusion of top of firing arm upright above timing disc less than min. allowable (Inspection to be performed on a 100% basis.)	Gage

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4.3.2.30 Body, movement, and lower cap assembly, F10542849 (prior to insertion of welsh plug) covering a detail of drawing D10542845

Defect	Method of Inspection
CRITICAL	
1. Firing arm not in the unfired position	Visual
2. Hammerspring assembly does not engage lug at Muzzle Action setting	Visual
MAJOR AQL = 0.40%	
101. Improper or incomplete assembly	Visual
102. Body assembly screws inadequately staked	Visual
*103. MA and zero line not in alignment prior to zero set operation	Visual
*104. Housing and plunger assemblies not engaged when set on Muzzle Action	Manual/Visual
*105. Zero setting incorrect (after tightening adjusting stud)	Visual
*106. Adjusting stud insecure after tightening screw	Visual
*107. Hammerspring assembly engages lug at 200 meter setting	Visual
*108. Adjusting stud not in timing disc tab slot	Visual
*109. Timing disc tab not in contact with shoulder of adjusting stud.	Visual
*Inspection to be performed on a 100% basis	

4.3.2.31 Fuze Assembly, D10542845 less detonator holder and Interrupter Assemb ly covering a detail of drawing D10542845

CRITICAL	
None Defined	
MAJOR AQL = 0.40%	
101. Detonator screw - above flush of body	Visual
102. Detonator screw assembly inserted improperly (Red end visible - improper)	Visual
103. Relay extends beyond flush	Visual
104. Relay improperly inserted in closing plug (Closed end not visible)	Visual
*105. Spinning or crimping of relay inadequate	Visual
106. Staking of closing plug assembly screws inadequate	Visual
107. Foreign matter	Visual
*100% inspection	

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4.3.2.32 Fuze Assembly, drawing D10542845

Defect	Method of Inspection
CRITICAL	
1. Interrupter in armed or partially armed position (check immediately prior to assembly into fuze see 6.4.1 and 6.4.2)	Visual
MAJOR AQL = 0.40%	
101. Fuze not set on (MA) muzzle action. (MA line on lower cap not in line with zero line on body) to be checked immediately prior to packaging	Visual
102. Total length exceeds max.	Gage
103. Interrupter not flush or below surface A	Visual
104. Length to flange exceeds max.	Gage
105. Improper or incomplete assembly	Visual
106. Threads damaged	Visual
107. Sealant missing	Visual
108. Marking missing, incorrect or illegible	Visual
109. Dust seal missing or not coated	Visual
MAJOR 100% (reference dwg 10542849)	
*110. Adjusting stud not engaged in timing disc tab slot	Visual
*111. Timing disc tab not contacting adjusting stud shoulder	Visual
*112. Zero setting gap incorrect at 100 meter setting	Gage
*To be checked through welsh plug holes after head, closing plug, detonator holder, and interrupter assembly have been secured and markings applied. (Rejected fuzes shall be resubmitted for a recheck of the tab engagement (Major 110 and 111) and zero setting gap (major 112) after rework.)	
113. Lower Cap not locked in position when set at MA (to be checked after checking zero set gap)	Manual/Visual

4.3.2.33 Packaging, Packing, and Marking.- The number of specimens and acceptance criteria for packaging, packing, and marking shall be in accordance with MIL-P-60412.

4.3.2.34 Process control audit.- Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings. Bending adjustment of the firing arm upright is not permitted at any time.

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4.3.2.34 (cont'd)

a. Laminated plates No. 1 Assembly, Drawing D10542811

(1) Staking the regulator

b. Movement Sub-Assembly, Drawing F10542817

(1) Application of lubricant
(2) Staking of bottom assembly screws (prior to regulation)
(3) Mainspring pre-winding operation (3 turns)

c. Movement Assembly, Drawing D10542813

(1) Regulation operation
(2) Movement rewound after regulation
(3) Backlash removal
(4) Locknut torqued prior to timing disc torque test
(5) Final lubrication

d. Body, Movement, and Lower Cap Assembly, Drawing F10542849

(1) Lubrication and insertion of "O" ring
(2) Lock ring torque operation
(3) Drilling operation for lock rings
(4) Application of sealant to screws and welsh plugs

e. Fuze Assembly, Drawing D10542845

(1) Head torque operation
(2) Insertion of dessicant pellet and cushion in closing plug loading assembly
(3) Staking of assembly screws
(4) Application of sealant
(5) Application of lubricant

f. Timing Disc Support Assembly, Drawing D10535688

(1) Spot welding

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4

4.3.3.1 Protective finish.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable standards or specifications listed on the drawing. (see 4.4.1)

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4.3.3.2 Hardness (Destructive).- Twenty samples from each lot of the following components shall be subjected to the hardness test. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot. Where plating is required, the sample parts submitted shall be unplated (see 4.4.2)

Component	Drawing No.
Gear, Segment Setting	D10535748
Spring, Hammer	C10543231
Gear, Escapement	D10542765
Pinion, Escapement	D9205044
Pinion, No. 1	C10543226
Pinion, No. 2	D10543223
Pinion, No. 3	C10543224
Pinion, No. 4	D10543225
Collar	C9205040
Arbor	D9205039
Lever, Escapement	F9205053
Spring, Timing Disc (see 4.3.3.3)	C9205083
Spring, Escapement	B9205074
Washer, Bearing	B9205130
Pin, Firing	C10521154
Washer, Timing Disc	B9205085
Housing (Non-destructive - see 4.4.2)	F9205032

4.3.3.3 Load,Timing Disc Spring (Destructive).- The same 20 springs subjected to the hardness test shall be subjected to the load specified on drawing C9205083. Failure of any spring to comply with the requirement shall be cause for rejection of the lot (see 4.4.3)

4.3.3.4 Compression,Firing Pin Spring (non-destructive).- Twenty from each lot shall be subjected to the compression test specified on drawing C10521132. Failure of any spring to comply with the requirement shall be cause for rejection of the lot (see 4.4.4).

4.3.3.5 Firing Arm Bend (Destructive)- Twenty (20) firing arms from each lot shall be subjected to the bend test as specified on drawing D9205090. Failure of any firing arm to comply with the requirements shall be cause for rejection of the lot (see 4.4.5)

4.3.3.6 Laminated Plate No. 1 Assembly

4.3.3.6.1 Push or Pull Test-Laminated Plate No. 1 Assembly (non-destructive).- Twenty-five (25) laminated plate no. 1 assemblies from each lot shall be subjected to the push of pull test specified on drawing D10542811. Failure of any plate to comply with the requirement shall be cause for rejection of the lot (see 4.4.6.1)

4.3.3.6.2 Torque (Non-destructive).- Each laminated plate no. 1 assembly produced shall be subjected to the torque test specified on drawing D10542811. Failure to comply with the requirement shall be cause for rejection of the laminated plate no. 1 assembly (see 4.4.6.2)

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4.3.3.7 Static load and torque

4.3.3.7.1 Static load.- Twenty-five (25) samples from each lot of the following sub-assemblies shall be subjected to the static load test. Failure of any sub-assembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of sub-assemblies as applicable (see 4.4.7 and 6.7)

Static Load	Drawing No.
Escapement Gear & Pinion Assembly	C9205045
Escapement Lever Assembly	D9205056
Firing Arm Assembly	C9212053
No. 1 Gear & Pinion Assembly	C10543220
No. 2 Gear & Pinion Assembly	C10520769
No. 3 Gear & Pinion Assembly	C10543222
No. 4 Gear & Pinion Assembly	C10543221
Pallet Lock Assembly (Pin)	C9205066
Arbor and Housing Assembly (Pins & Lug) (Non-Destructive)	D10542800
#5 Gear 9205162 & Arbor Assembly 9205042 (Destructive)	

4.3.3.7.2 Torque.- Twenty-five (25) samples from each lot of the following sub-assemblies shall be subjected to the torque test. Failure of any sub-assembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of sub-assemblies, as applicable (see 4.4.7 and 6.7)

Sub-Assembly	Drawing No.
Escapement Gear & Pinion Assembly	C9205045
No. 1 Gear & Pinion Assembly	C10543220
No. 2 Gear & Pinion Assembly	C10520769
No. 3 Gear & Pinion Assembly	C10543222
No. 4 Gear & Pinion Assembly	C10543221
Escapement Lever Assembly	D9205056
*Arbor and Housing Assembly	D10542800 (Reference only)

*#5 Gear 9205162 and Arbor Assembly 9205042 (Destructive)

*Gear 9205162 and Arbor Assembly 9205042 only shall be assembled for static load and torque test sample using the same assembly methods and equipment that shall be used in production of Arbor and Housing assembly.

4.3.3.8 Hammerspring Deflection (Non-Destructive).- The hammerspring of each housing and hammerspring assembly produced shall be subjected to the load as specified on drawing C10543235. Failure to comply with the requirements shall be cause for rejection of the assembly (see 4.4.8).

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4.3.3.9 Kinetic Energy of Firing Pin (Non-Destructive).- Fifty (50) movement sub-assemblies from each lot shall be tested to determine compliance with the requirement specified on drawing F10542817. Failure of any assembly to comply with the requirement shall be cause for rejection of the lot (see 4.4.9).

4.3.3.10 Timing Disc Torque (Non-Destructive).- The timing disc of each movement assembly produced shall be subjected to the torque specified on drawing D10542813. Failure to comply with the requirement shall be cause for rejection of the movement assembly (see 4.4.10).

4.3.3.11 Movement Sub-Assembly

4.3.3.11.1 Centrifugal Weights - Function - Non-destructive).- The centrifugal weights within each no. 7 plate assembly produced shall be tested to determine compliance with the function requirement of drawing F10542817. Failure of the weights to separate sufficiently to clear the slot in the plate shall be cause for rejection of the movement sub-assembly. (see 4.4.11.1)

4.3.3.11.2 Centrifugal Weights - Non-Function - (Non-destructive).- The centrifugal weights within each no. 7 plate assembly produced shall be tested to determine compliance with the non-function requirement of drawing F10542817. Failure of either centrifugal weight to remain in the fully locked position shall be cause for rejection of the movement sub-assembly (see 4.4.11.2).

4.3.3.11.3 Static function test without oil on the escapement pallet and escapement gear teeth (non-destructive).- Each movement sub-assembly produced shall be subjected to a static function test without the application of oil, where specified on Drawing F10542817 (Note 14). Failure of the movement to run from the fully wound position to a fully unwound position shall be cause for rejection of the movement sub-assembly (see 4.4.11.3).

4.3.3.11.4 Spin function and fire (Non-destructive).- Each movement sub-assembly produced shall be tested to determine compliance with the operation spin requirement as specified in paragraph 3.2.2 and while spinning the unit shall fire. Failure to comply with the spin requirement or failure of the firing arm upright to enter the timing disc slot and release the firing pin shall be cause for rejection of the movement sub-assembly (see 4.4.11.4).

4.3.3.11.5 Load-Setback weights (Non-destructive).- The setback weights within each movement sub-assembly shall be subjected to the load as specified on drawing F10542817. Failure of any setback weight to comply with the requirement shall be cause for rejection of the movement sub-assembly (see 4.4.11.5).

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4.3.3.12 Movement Assembly

4.3.3.12.1 Static regulation (Non-destructive).- The regulation tape record (9 seconds minimum) of each movement produced shall be checked for compliance with the requirement of 3.2.1. Any movement which fails to comply shall be rejected (see 4.4.12.1 and 6.8).

4.3.3.12.2 Spin - Non-function (Non-destructive). - The pallet lock within each movement assembly produced shall be tested to determine compliance with the non-function requirement of drawing D10542813. Failure to comply with the requirement shall be cause for rejection of the movement assembly (see 4.4.12.2).

4.3.3.12.3 After backlash removal operation (Non-destructive).- Each movement assembly produced shall be tested to determine compliance with the torque requirement of drawing D10542813, Note 9. Any visible motion of the timing disc when the specified torque is applied shall be cause for rejection of the movement assembly (see 4.4.12.3)

4.3.3.13 Body, Movement, and Lower Cap Assembly

4.3.3.13.1 Adjusting stud security (Non-destructive).- After zero setting operation, each assembly produced shall be tested to determine security of the adjusting stud by application of the specified torque on drawing F10542849. Any assembly that fails to comply shall be rejected (see 4.4.13.1)

4.3.3.14 Detonator Holder and Interrupter Assembly

4.3.3.14.1 Arming (Non-destructive).- The interrupters within each assembly produced shall be tested to determine compliance with the arming requirement of drawing C10533945. Failure to comply with the requirement shall be cause for rejection of the assembly (see 4.4.14.1 and 6.4.1)

4.3.3.14.2 Non-Arming (Non-destructive).- The interrupters within each assembly produced shall be tested to determine compliance with the non-arming requirement of drawing C10533945. Failure to comply with the requirement shall be cause for rejection of the assembly (see 6.4.3 and 4.4.14.2).

4.3.3.15 Fuze Assembly

4.3.3.15.1 Setting torque (Non-destructive).- Each fuze assembly produced shall be tested to determine compliance with the setting torque requirement of drawing D10542845. Failure to comply with the requirement shall be cause for rejection of the fuze assembly (see 4.4.15.1).

4.3.3.15.2 Torque - Housing and Plunger Assembly - (Non-destructive).- Each fuze assembly produced shall be tested to the specified torque on drawing D10542845. Failure of the lower cap to remain at the meter setting shall be cause for rejection of the fuze assembly (see 4.4.15.2).

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4.3.3.16 Environmental tests (lots - destructive).- Twenty-two (22) assemblies from each production lot shall be subjected to the following MIL-STD tests, with the fuze set on muzzle action "MA".

Test	Quantity	Requirement	Criteria	Method
Jolt	4	3.3.1	4.2.5.1	4.4.16.1
Jumble	4	3.3.2	4.2.5.1	4.4.16.2
Forty Foot Drop	5	3.3.3	4.2.5.1	4.4.16.3
Waterproofness	4	3.3.4	4.2.5.1	4.4.16.4
Transportation-Vibration	5	3.3.5	4.2.5.1	4.4.16.5

4.3.3.17 Ballistics (see 4.4.17 Destructive).- Sixty (60) fuzes from each lot shall be subjected to the following ballistic test phases:

Weapon	Phase (Ambient Temp)	Quantity	Requirement	Method
90 mm	MA	20	3.4.1	4.4.17.1
90 mm	400 meters	20	3.4.2	4.4.17.2
90 mm	3000 meters	20	3.4.2	4.4.17.2

4.3.3.17.1 Acceptance Criteria.- The occurrence of any of the following shall be cause for rejection of the lot the ballistic sample represents:

a. More than one prior function in the combined distance function phases (see 6.2.1 and 6.2.2)

b. Failure of more than one round to function within 50 feet in the Muzzle Action phase.

c. Any of the following in the distance function phases:

(1) More than two duds or two outliers - offtimes or one dud and one outlier - offtime in the combined phases.

(2) Failure to meet the time limits in Table I by an estimated lot percent defective greater than the appropriate value given in Table II for each distance function phase (see 4.4.17.2 and 6.2.4)."

TABLE II

Sample Size	Acceptable % Defective
20	4.09
19	4.10
18	4.12
17	4.16
16	4.22
15	4.31

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4.3.4 Inspection equipment.- The examinations and tests shall be made using equipment listed in IEL-10542845. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test Methods and Procedures

4.4.1 Protective finish.- The test procedures for protective finish shall be in accordance with the applicable specification referenced on the component drawing.

4.4.2 Hardness.- Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three hardness readings taken on each sample unit shall be used as the individual value. Housing F9205032 hardness test shall be made on the bottom surface. Housing must be supported by shoulder to prevent deflection. Readings to be taken 0.020 inch min from major diameter.

4.4.3 Load - Timing Disc Spring.- The test shall be performed using the values as specified on drawing C9205083. The timing disc springs that have been subjected to the load test shall be scrapped.

4.4.4 Compression.- The firing pin springs shall be compressed for the length of time specified on drawing C10521132

4.4.5 Firing arm bend.- The test shall be performed as specified on drawing C9205090. The firing arms that have been subjected to the bend test shall be scrapped.

4.4.6 Laminated Plate No. 1 Assembly

4.4.6.1 Push or Pull Test.- The push shall be applied as indicated using the values as specified on drawing D10542811.

4.4.6.2 Torque.- The test shall be performed after completion of the staking operation using the values as specified on drawing D10542811.

4.4.7 Static Load and Torque.- The test shall be performed using the values as specified on the applicable drawing. All #5 gear and arbor assembly samples tested shall be scrapped.

4.4.8 Hammerspring Deflection - Housing and Hammerspring Assembly.- The required load shall be applied at the specified location parallel to the vertical axis of the assembly as specified on drawing C10543235. The hammerspring must return to surface "C" when released.

4.4.9 Kinetic Energy of Firing Pin - Movement Sub-Assembly.- The assembly shall be placed in a fixture and tested for kinetic energy by releasing the firing pin from the cocked position.

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4.4.10 Timing Disc Torque.- The torque test shall be performed after assembly of the timing disc assembly to the movement and assembly of the locknut on the arbor. The highest and lowest readings will be taken in one uninterrupted complete revolution of the timing disc for determination of compliance with the requirements of drawing D10542813.

4.4.11 Movement Sub-Assembly

4.4.11.1 Centrifugal Weights - Function.- The spin function test shall be performed with the setback weights depressed or omitted. The assembly shall be rotated at the R.P.M. specified on drawing F10542817 in the counterclockwise direction when viewed from the top, (using a strobe) for a period of not more than 5 seconds at constant R.P.M. to determine compliance with the requirement (see 6.5.1)

4.4.11.2 Centrifugal Weights - Non-function.- The spin non-function test shall be performed with the setback weights depressed or omitted. The assembly shall be rotated at the R.P.M. specified on drawing F10542817 in the counterclockwise direction when viewed from the top, (using a strobe) for a period of not less than five seconds at constant R.P.M. to determine compliance with the requirement (see 6.5.2).

4.4.11.3 Function test without oil on the escapement pallet and escapement gear teeth.- Upon manual release of the lever lock slide and starter the function shall be timed for nine (9) seconds, min. This test shall be performed prior to the spin function and fire test.

4.4.11.4 Spin function and fire - movement sub-assembly less timing disc support setback weights and utilizing a simulated timing disc.- The high spin function and fire shall be performed on a fully would movement sub-assembly with a simulated timing disc. The movement sub-assembly shall be rotated at the R.P.M. specified in paragraph 3.2.2 in the clockwise direction when viewed from the bottom for a period of not less than nine (9) seconds at constant R.P.M. Compliance with the requirement shall be determined after the test by the position of the simulated timing disc and that the firing arm upright entered the timing disc slot and released the firing pin. The simulated timing disc design shall be approved by Q3300, Frankford Arsenal.

4.4.11.5 Load-Setback Weights.- The test shall be performed as outlined using the values as specified on drawing F10542817.

4.4.12 Movement Assembly

4.4.12.1 Regulation - Movement Assembly with timing disc assembly, timing disc support, and setback weights.- The movement assembly shall be regulated using the operational procedure furnished with the equipment. Timekeeping and performance shall be checked for a period of not less than nine (9) seconds. No adjustment shall be made on the movement during the nine (9) second check runs. A regulation tape record shall be obtained for each movement assembly.

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4.4.12.2 Spin Non-Function.- The spin non-function test shall be performed on a completely wound movement assembly and after regulation has been performed on the movement. The movement assembly shall be rotated at the R.P.M. specified on the movement sub-assembly. Drawing F10542817 in the clockwise direction when viewed from the bottom, (using a strobe) for a period of not less than 5 seconds at constant R.P.M. This shall be the last spin test that the movement is subjected to.

4.4.12.3 Backlash Removal.- The specified torque shall be applied to the lock nut in a clockwise direction. Care must be exercised in order than the maximum torque specified is not exceeded. This test should be performed after the last wind.

4.4.13 Body, Movement, and Lower Cap Assembly

4.4.13.1 Adjusting Stud Security - Body, Movement, and Lower Cap Assembly.- The torque specified on drawing 10542849 shall be applied in clockwise direction. Visible loosening shall be determined by the change in zero set gap.

4.4.14 Detonator Holder and Interrupter Assembly

4.4.14.1 Arming.- The unarmed interrupter assembly shall be rotated in a counterclockwise direction when viewed from interrupter cover end, at the rpm specified on drawing C10533945. The arming test shall be performed prior to the non-arming test. At the conclusion of the test, the interrupter shall be returned to the unarmed position (see 6.4.1).

4.4.14.2 Non-Arming.- The unarmed interrupter assembly (see 6.4.3) shall be rotated in a counterclockwise direction when viewed from the interrupter cover end, at the rpm specified on drawing C10533945. The

4.4.15 Fuze Assembly

4.4.15.1 Setting Torque.- The test shall be performed using the values as specified on the applicable drawing. The highest and lowest readings will be taken in one uninterrupted complete revolution of the lower cap for determination of compliance with the requirement of drawing D10542845.

4.4.15.2 Torque - Housing and Plunger Assembly.- The test shall be performed using the specified torque at each setting as specified on drawing D10542845.

4.4.16 Environmental tests.- The procedures used shall be as set forth in the following sub-paragraphs, as applicable. Where spin testing is required, the non-arming or non-function test shall be performed prior to the arming or function test. The components of the disassembled fuzes shall be scrapped. All adverse conditions observed shall be reported to the responsible engineering element.

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4.4.16.1 Jolt.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 101 with the fuze orientations in the following sequence:

a. Nose down

b. Horizontal

c. Nose up

4.4.16.1.1 The fuze shall be disassembled and inspected as follows:

a. Prior to disassembly, check position of "MA" line relative to zero line. Check for security of the housing and plunger assembly (lower cap moves).

b. Remove head, (holding lower cap securely)

c. Remove lower cap and check engagement of setting pin with timing disc lug, check centrifugal weights (both closed).

d. Remove detonator holder and interrupter assembly and check position of interrupter slides.

e. Remove closing plug and check relay.

f. Remove movement from body. Check position of firing pin, firing arm upright, pallet lock slide and escapement lever upright. Check timing disc torque and for separation of movement plates.

g. Spin detonator holder and interrupter assembly for non-arming and arming.

h. Disassemble major sub-assemblies to the extent that all critical parts can be examined, such as gear and pinion assemblies, etc., for security of assembly, broken or bent pivots and teeth, fractured or bent firing arm or shaft or other part which might allow accidental release of firing pin. Check condition of all explosive elements.

4.4.16.2 Jumble.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 102. Then fuze shall be disassembled and inspected as outlined in 4.4.16.1.

4.4.16.3 Forty Foot Drop.- The test shall be performed in accordance with the procedure specified in MIL-STD-33, Test No. 103. The fuze shall be disassembled and inspected as follows.

a. Remove detonator holder and interrupter assembly and check position of interrupter slides.

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4.4.16.3 (cont'd)

b. Remove interrupter cover and check condition of parts (where slides are found in a partially armed or fully armed position)

c. Check condition of all explosive elements.

4.4.16.4 Waterproofness.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 108, except as provided in 3.3.4. The fuze shall be tested with all vulnerable areas exposed. The fuze shall be disassembled and inspected as follows:

a. Remove head, detonator holder, and interrupter assembly, lower cap, check all visible internal surfaces for evidence of water entry.

4.4.16.5 Transportation - Vibration.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 104, with the fuze orientations in the following sequence:

a. Parallel to longitudinal axis, nose up

b. First transverse axis to be parallel to a line running through the marking "MA".

c. Second transverse axis to be parallel to a line running through the marking 23 (approximate).

4.4.16.5.1 The fuze shall be disassembled and inspected as follows:

a. Prior to disassembly, check position of "MA" line relative to zero line.

b. Remove welsh plugs and check engagement of setting pin with timing disc lug.

c. Check for security of the housing and plunger assembly (lower cap moves).

d. Remove head (holding lower cap securely)

e. Remove lower cap and check centrifugal weight (both closed)

f. Remove detonator holder and interrupter assembly and check position of interrupter slides. Spin assembly for non-function and function.

g. Remove closing plug and check relay.

h. Remove movement from body. Check position of pallet lock slide and escapement lever upright. Check timing disc torque. Spin movement for non-function and function.

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4.4.16.5 (cont'd)

i. Remove timing disc retainer support and check setback weights. Spin movement for centrifugal weight function.

j. Disassemble major sub-assemblies to the extent that all critical parts can be examined, such as gear and pinion assemblies, etc., for broken or bent pivots and teeth, fractured or bent firing arm or shaft or other parts which might allow accidental release of firing pin. Check condition of all explosive elements.

4.4.17 Ballistics.- The fuze assembly, drawing F10542845 shall be assembled to the appropriate cartridge and fired as follows:

4.4.17.1 Muzzle Action.- The sample fuzes shall be fired as specified in 3.4.1 under the conditions delineated in 4.2.6 or 4.3.3.17, as appropriate. The rounds to be fired at extreme temperature shall be conditioned as provided in 4.4.17.3.1. The function distance from the muzzle of the weapon shall be measured and recorded.

4.4.17.2 Distance function.- The sample fuzes shall be fired as specified in 3.4.2 under the conditions delineated in 4.2.6 or 4.3.3.17, as appropriate. The rounds to be fired at extreme temperatures shall be conditioned as provided in 4.4.17.3.1. The time-to-function of each fuze shall be measured to the nearest millisecond by using two independent fuze chronograph systems. The time-to-function shall also be determined by using a Hawk-Doppler velocimeter system. All measurements shall be recorded.

4.4.17.2.1 Lot Percent Defective.- The time values obtained by the chronograph systems shall be used to calculate the estimate of lot percent defective. When both chronograph systems provide a time value for an individual fuze, the average shall be taken as the individual fuze value. Individual fuze values shall be rounded off to the second decimal place using the method prescribed by the American Society for Testing Materials. Where both chronograph systems fail to provide a value, the value obtained by the velocimeter system shall be used. The estimate of lot percent defective shall be calculated using the method prescribed in MIL-STD-414 Example B3. For sample sizes 16 through 19 the estimated lot percent defective value shall be determined by interpolation between the values given in Table B5, MIL-STD-414 for sample sizes 15 and 20.

All appropriate time values obtained for each distance function phase shall be used in the initial calculation of sample average ($\bar{x}$) and standard deviation (s). The standard deviation value obtained shall then be used to determine statistical outliers (non-normal values) by using the criteria $\bar{x}$ plus or minus 2.88 times the standard deviation. A value judged to be an outlier shall be excluded and the remaining values used to recalculate $\bar{x}$ and s. A minimum of 15 statistically normal time values are required for each phase (see 6.2.4).

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4.4.17.3 Extreme Temperature Tests.- The sample fuzes, shall be fired as specified in 3.4.3 and 3.4.4 under the conditions delineated in 4.2.6 time-to-burst shall be recorded for information purposes.

4.4.17.3.1 Temperature conditioning (-40°F and +140°F).- The test chamber shall be at ambient temperature at the start of the test and the chamber maintained at a temperature of minus 40°F or plus 140°F as applicable, until stabilization of the fuze temperature is reached (see 6.3). After stabilization, the fuze shall be saturated at minus 40°F or plus 140°F as applicable, for one hour, minimum.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.- Preservation and packaging shall be Level "A".

5.1.1 Level "A".- Packaging shall be in accordance with Drawing F88864492, except air pressure test is not required.

5.1.2 Level "B".- Packing shall be in accordance with Drawing C8886458.

5.1.3 Marking.- Ammunition box shall be marked in accordance with Drawing F8864492. Wood box shall be marked in accordance with Drawing C8886458.

5.2 Level "C".- Unit boxes shall be packed in accordance with Drawing C9214704 and C9214705.

5.3 The ICC marking shall be "Percussion Fuzes-Handle Carefully."

6. DEFINITIONS AND NOTES

6.1 Dimensions

6.1.1 Compliance with tests (proving ground, ballistic performance, and shop) required by the specifications and drawings is mandatory. The contractor will not assume, nor does the government guarantee, that all possible combinations permitted by the tolerance limits of the specifications and drawings will consistently satisfy the test requirements. Therefore, the manufacturer is obligated to choose those combinations of tolerances and fits within the limits of the specifications and drawings that best suit his process needs and still satisfy the testing requirements.

6.1.2 Dimensions which are specifically referred to in the classification of defects, as listed in Section 4, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate fuze performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on fuze functionability. This relaxation

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of requirements applies to "after the fact" deviations only, to dimensions only, and does not constitute authority to establish new dimensions or to relax efforts to achieve the required sizes within the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

6.1.3 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Ballistic Parameters

6.2.1 Prior Function.- Shall be defined as the occurrence of any of the following:

a. The round detonates in the bore of the gun except when fired MA

b. The round detonates before 0.117 seconds when set at 400 meters

c. The round detonates before 3.271 seconds when set at 3000 meters

6.2.2 Outlier - Offtime.- An outlier - offtime is a statistical outlier (non-normal value) which falls beyond the time limits specified in Table I (includes prior function).

6.2.3 Dud.- A dud is considered to have occurred if the round fails to detonate during flight.

6.2.4 Where the number of statistically normal time values obtained in either distance function phase is less than 15 and the lot is not rejected because of other reasons, the disposition of the lot shall be referred to the responsible procuring activity. The Government reserves the right to require the contractor to submit additional test samples. The number of additional test samples shall be sufficient to obtain a combined total of values in order that the occurrence of outliers will not reduce the number of normal values to less than the minimum required. All normal values obtained in the combined tests shall be used to determine lot acceptance or rejection. Attribute characteristic failures, e.g. duds occurring in the additional samples shall not be used to reject the lot.

6.3 Temperature stabilization of fuzes may be checked by a thermal measuring device in good thermal contact with the largest centrally located internal mass.

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6.4 Armed/Non-Armed Interrupter

6.4.1 Armed.- Both interrupter slides are out of position so that face of detonator is exposed in the cover hole.

6.4.2 Partially armed.- One interrupter slide has remained in the outward position.

6.4.3 Non-Armed.- Both interrupter slides cover the hole in the plate.

6.5 Function/Non-Function - Centrifugal Weights

6.5.1 Function.- Centrifugal weights must be opened when the arming speed is attained to allow the firing arm to enter between the weights and travel freely to the fire position.

6.5.2 Non-Function.- Centrifugal weights remain in place and prevent the firing arm upright from moving in.

6.6 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6.7 Screening of Lots.- Caution shall be exercised so no damage occurs to components during static load and torque testing.

6.8 Regulation.- Regulation tape record identified with its respective movement assembly shall be forwarded to Commanding Officer, Frankford Arsenal, ATTN: SMUFA-J6200.

6.9 Defective Fuzes.- Where sample fuzes are found to be defective as received at the ballistic test facility, all tests of the lot sample shall be held in abeyance pending investigation by the responsible procuring activity.

6.10 Order Data.- Procurement documents should specify the following:

a. Title, Number, and Date of this specification

b. Level of packing required (see 5)

Custodian:

Army - MU(FA)

Preparing Activity:

Army - MU(FA)
Project No. 1390-A816

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